



WORKSHOP REPORT

Workshop Title:

Arduino-Based Drone Assembly, Simulation & Flight Training

Organized By:

Flytutor Technologies Private Limited

Student Details:

Student Name : _____
Year : _____
Department : _____
College : _____
Date : _____

Introduction:

The workshop on Arduino-based drone development was conducted to provide practical knowledge about drone technology, including assembly, simulation, and flight operations. This workshop helped in understanding the working principles of drones and their real-world applications.

Day 1: Drone Assembly & Simulation

Session Details :

On the first day, the session started with an introduction to drones and their applications such as agriculture, surveillance, mapping, and delivery systems. The trainers explained the basic components required to build a drone.

Topics Covered :

Introduction to UAVs and drone types
Overview of Arduino as a flight controller
Components: BLDC motors, ESCs, propellers, Li-Po battery
IMU sensors (Gyroscope & Accelerometer)
Transmitter and receiver

Practical Session :

Assembling the drone frame
Mounting motors and ESCs
Wiring connections with Arduino
Sensor integration

Simulation Training :

Introduction to drone simulator
Understanding controls: throttle, pitch, roll, yaw
Practicing virtual flying

Outcome of Day 1 :

By the end of Day 1, I was able to understand the structure of a drone, assemble its components, and practice basic flight controls using a simulator.

Day 2: Flight Training & Calibration**Session Details :**

The second day focused on configuring the drone and performing real-time flight operations.

Topics Covered :

Firmware uploading to Arduino
ESC calibration
Sensor calibration
Transmitter binding

Pre-Flight Checklist :

Battery voltage check
Propeller direction verification
Motor testing
Safety precautions

Practical Flight Training :

Drone arming and takeoff
Hovering practice
Directional movements
Landing techniques

Outcome of Day 2 :

By the end of Day 2, I was able to perform basic drone flying operations and understood the importance of calibration and safety checks.

Learning Outcomes :

Gained knowledge about drone components and working
Learned drone assembly and wiring
Understood simulation-based training
Acquired basic drone flying skills
Learned safety measures and troubleshooting

Conclusion :

The workshop was highly informative and provided hands-on experience in drone technology. It enhanced my practical skills and gave me a clear understanding of drone assembly and flight operations. This workshop will be beneficial for future learning and career opportunities in the drone industry.



